Topo4D++: Realistic Physically Based 4D Head Capture With Topology-Preserving Gaussian Splatting and Expression Priors

Yuhao Cheng, Xuanchen Li, Xingyu Ren, Jianyu Wang, Haozhe Jia, Di Xu, Wenhan Zhu, Bingbing Ni, and Yichao Yan

***Abstract*—4D head capture aims to generate dynamic facial meshes in the same topology with corresponding UV maps, which requires temporal correspondence between 3D head models. Existing pipelines either involve manual processing of artists or employ constraints such as landmark tracking and optical flow, failing to achieve a trade-off between accuracy and efficiency. To enhance this process, we propose Topo4D++, a novel framework for automatic geometry and texture reconstruction that optimizes densely aligned 4D heads and 8 K BRDF maps directly from calibrated multi-view videos. Our key insight is to represent facial models as a set of dynamic 3D Gaussians with fixed topology, where the Gaussian centers are bound to the mesh vertices. This enables tracking all vertices rather than sparse vertices on the face accurately by leveraging the inverse rendering capabilities of 3D Gaussian Splatting (3DGS), while also enabling ultra-high-resolution texture generation. To maintain face structure during dynamic 3DGS optimization, we propose to optimize geometry and texture alternatively under physical and topological constraints frame-by-frame and employ blendshape-based expression priors to address extreme expressions. Then, we propose to extract dynamic facial meshes in a regular wiring arrangement and high-fidelity textures with pore-level details from the learned Gaussians. Finally, we train a diffusion-based model to generate BRDF texture maps to achieve physically based rendering. Given the absence of a universal benchmark, we construct JHead, a novel benchmark for the comprehensive evaluation of 4D head capture methods. Extensive experiments on different datasets demonstrate that our method is generalized to different capture systems, identities, and expressions, outperforming current state-of-the-art head reconstruction methods in both mesh and texture qualitatively and quantitatively.**

***Index Terms*—4D head reconstruction, high-resolution BRDF texture generation, 3D Gaussian splatting.**

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Yuhao Cheng, Xuanchen Li, Xingyu Ren, Jianyu Wang, Bingbing Ni, and Yichao Yan are with the MoE Key Lab of Artificial Intelligence, AI Institute, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: chengyuhao@sjtu.edu.cn; lixc6486@sjtu.edu.cn; rxy_sjtu@sjtu.edu.cn; ppjiong11@sjtu.edu.cn; nibingbing@sjtu.edu.cn; yanyichao@sjtu.edu.cn).

Haozhe Jia and Di Xu are with Huawei Cloud Computing Technologies Company, Ltd., Guiyang 550029, China (e-mail: jiahaozhe1@huawei.com; xudi21@huawei.com).

Wenhan Zhu is with Xueshen AI, Shanghai 200052, China (e-mail: whzhu@foxmail.com).

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I. Introduction

4D HEAD capture aims to acquire temporally continuous facial assets in a fixed topology, including meshes and textures. It is widely utilized in entertainment media, such as computer games, movies, and interactive AR/VR, enabling the creation of dynamic and photo-realistic facial animations that enhance immersion and visual fidelity. Creating captivating and lively 4D facial assets requires addressing the following key challenges: **1)** faithfully representing faces with fixed **topology** and UV, **2)** maintaining **temporal stability** between different frames, especially in extreme expressions, **3)** reconstructing high-fidelity geometric details and high-quality Bidirectional Reflectance Distribution Function (BRDF) texture maps, and **4)** ensuring the efficiency of the framework, considering the runtime and degree of automation.

The industrial pipeline typically employs professional equipment, e.g., Light Stage [1], to capture high-quality multi-view videos. Then, multi-view stereo (MVS) [2], [3] is used to compute the facial scans of each frame, followed by a non-rigid registration [4] process to warp the topologically aligned faces onto the scans. To maintain temporal consistency, hundreds of points are marked on the faces of the actors and are tracked for accurate non-rigid registration. Then, experienced artists manually correct the tracking results using professional software such as Wrap4D and Track4D [5] to ensure the alignment quality, which is labor-intensive and time-consuming. Therefore, there is an urgent demand to develop more automated workflows to accelerate the 4D facial asset reconstruction.

To automate the process, researchers have mainly developed two kinds of methods in the last few years, i.e., regression-based methods and optimization-based methods. Regression-based methods generally employ pre-trained models to generate **topological** facial assets frame-by-frame. Some works [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22] utilizes parametric models to fit the facial images for mesh creation, where parametric texture or inverse rendering is employed to generate the corresponding textures. These methods are highly efficient, but due to their limited expressive abilities, they struggle to produce high-quality facial models with diverse identities and complex expressions. Other regression-based methods [23], [24], [25], [26], [27], [28] propose to directly regress facial models from multi-view images,

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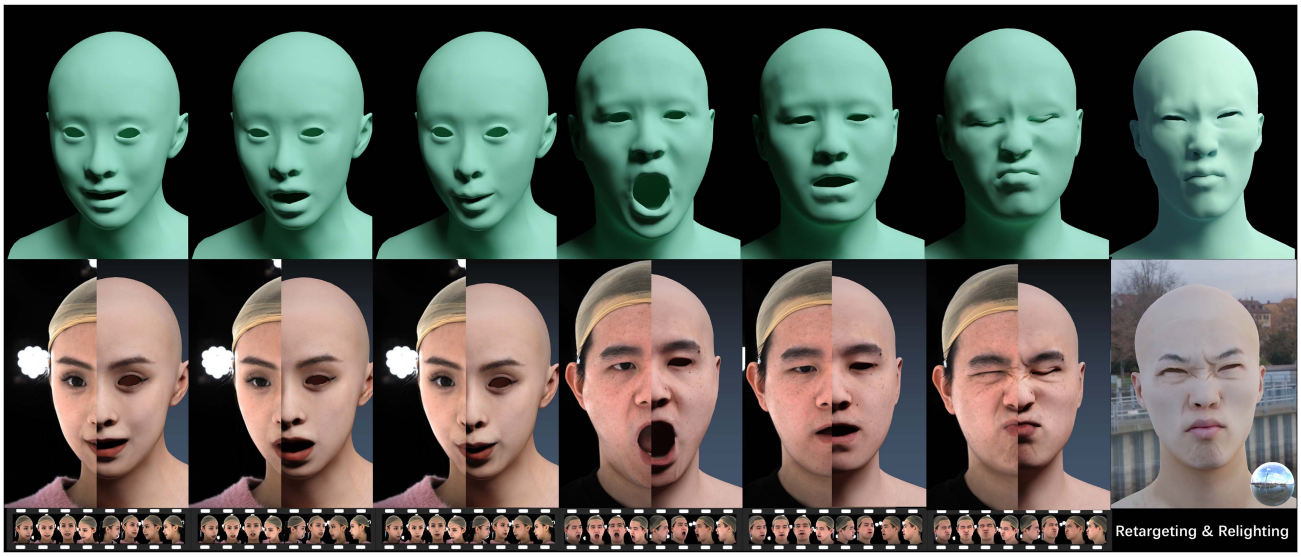


Fig. 1. Example results of our Topo4D++. Our method can produce temporally consistent topological head meshes with high-fidelity 8 K textures from calibrated multi-view videos. Captured 4D models can be applied to retargeting and relighting applications.

where a large amount of expensive pre-processed 3D models are utilized for training. Typically, these methods can capture consistent features across multiple views to predict accurate geometry, and several recent works [25], [28] can further generate high-resolution BRDF textures with a super-resolution module. Nevertheless, super-resolution modules often introduce artifacts and fail to faithfully recover face details. Furthermore, most of the regression-based methods are designed to reconstruct each frame individually without considering frame continuity, thereby struggling to maintain the temporal coherence between frames. Meanwhile, the optimization-based methods [29], [30], [31], [32] employ supervisions such as optical flow and facial landmarks to automatically warp template models within frames to acquire temporally stable assets. However, these additional supervisions become inaccurate in extreme expressions, thus failing to faithfully reconstruct heads. Additionally, they need careful parameter tuning for different subjects to achieve optimal results and require a large processing time. In summary, these methods fail to achieve a good trade-off between reconstruction accuracy, stability, and efficiency.

The recent progress in 3D Gaussian Splatting (3DGS) [33] and its advancements in 4D scene representation [34], [35], [36], [37], [38] bring us inspirations. These methods leverage point-based representations to model scenes and achieve **temporally consistent** reconstruction by tracking the movement of each point in the dynamic scenarios. Moreover, thanks to its advanced rendering pipeline, it can achieve ultra-high resolution texture learning and fast perspective rendering in a memory-efficient manner. Considering their success in 4D scene representation, it would be highly desirable if we could inject their stable tracking and efficient rendering properties into the 4D head capture framework. However, it is non-trivial due to 4 major challenges. First, 3DGS is a highly non-structured representation, which needs to be carefully aligned to the mesh and UV map to satisfy the requirements of head capture. Second, in the dynamic scenarios, it lacks physical constraints for the Gaussian points to move following the guidance of facial expressions, resulting in misalignment, especially in extreme expressions. Third, it is challenging to extract high-quality meshes and high-resolution texture maps with fixed topology directly from 3DGS. Finally, spherical harmonic lighting is utilized in vanilla 3DGS, which poses a further challenge to learn decoupled BRDF properties for physically based rendering.

To fully leverage the potential of 3DGS in 4D head capture and overcome the aforementioned challenges, we propose a novel framework, **Topo4D++**, to obtain high-fidelity 4D meshes and temporally stable textures, as shown in Fig. 1. **1)** To maintain the quality of photo-realistic rendering and extract meshes and textures with **fixed topology**, we improve 3DGS by explicitly linking each 3D Gaussian to each vertex of the pre-defined topological facial meshes, named Gaussian Mesh. Hence, it simultaneously obtains topology-constrained meshes and photo-realistic textures. Besides, inheriting the advantages of 3DGS, the optimization process is fast and memory-efficient. **2)** To obtain dynamic facial meshes, we propose to optimize the Gaussian Mesh frame-by-frame under physical and topological constraints to ensure temporally topological stability and regular mesh arrangement, meeting the high-quality requirements of facial assets. Blendshape-based expression prior is utilized to guide the movement of Gaussians for better initialization in extreme expressions. **3)** After optimization, high-quality meshes can be extracted by aligning the Gaussian surface with the rendered surface using a geometric normal prior. We also design the UV densification module to learn detailed geometric offsets and textures during the refinement process, so that we can obtain ultra-high-resolution texture and displacement maps with fine-grained pore-level details by inversely mapping the Gaussian colors and position offset into UV space. **4)** Finally, we train a diffusion-based conversion model to generate albedo

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and specular maps from texture maps to achieve physically based rendering.

Besides the Topo4D++ framework, we also observe a lack of benchmarks for evaluating 4D head capture performance. Therefore, we construct a novel benchmark JHead to provide a comprehensive and effective evaluation of 4D head capture methods. In the creation of JHead, we utilize our Light Stage system to capture the dynamic performances of 20 different actors. Then, we obtain the ground truths by employing professional software Agisoft Metashape [39] to reconstruct high-precision meshes from the calibrated multi-view images. The effectiveness of 4D head capture methods can be evaluated by measuring the distance between the reconstructed meshes and the ground truths.

The effectiveness of Topo4D++ has been validated by extensive experiments in our JHead and other open-source datasets, e.g., FaMos [24] and Multiface [40]. To our knowledge, Topo4D++ is the first method to reconstruct high-fidelity 4D facial models and 8 K texture maps with 3DGS, demonstrating the potential of 3DGS for dynamic face reconstruction and ultra-high-resolution tasks. The obtained meshes and textures can benefit facial asset generation and facial animation. In summary, our contributions include:

- We propose a novel framework, Topo4D++, for efficient reconstruction of high-quality 4D head meshes and 8 K photo-realistic BRDF textures with pore-level details from multi-view videos.
- We design a novel Gaussian mesh representation and multi-constraint optimization process that introduces physical, topological, and blendshape-based expression priors to obtain temporally stable facial models in the pre-defined topology and UV structure.
- We propose a UV densification method and a diffusion-based conversion model to learn the detailed geometric offsets and reflectance attributes, achieving high-fidelity BRDF maps generation.
- We establish a benchmark JHead to comprehensively evaluate the performance of 4D head capture methods, filling a gap in this field.

This work is an extension of Topo4D [41]. Compared to the conference version, Topo4D++ has the following new features: **1)** Expression-prior-based initialization method to better reconstruct heads in extreme expressions. **2)** Extending the UV densification method to optimize geometric offset to generate displacement maps, thus achieving highly-detailed geometry reconstruction. **3)** Diffusion-based conversion model to generate BRDF maps. **4)** An open-released large 4D head capture evaluation benchmark JHead. **5)** More comprehensive experiments and ablation studies.

II. Related Works

A. Registered Facial Model Acquisition

Acquiring high-fidelity facial models in pre-defined topological structures has been a long-standing research challenge. Beginning with 3D Morphable Models [42], many methods [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [19], [20] employ parametric models to achieve face reconstruction from single- and multi-view images. However, such methods struggle to faithfully reconstruct faces because the 3DMM representation lacks the necessary fine-grained details. Non-rigid ICP [4], [43], [44], [45] methods deform canonical models to fit scans reconstructed by MVS methods with high quality. However, naively extending these methods to 4D videos will result in temporal instability, leading to texture drift. To maintain temporal stability, high-precision models are registered frame-by-frame in existing CG pipelines using professional software, e.g., Wrap4D and Track4D [5], which demands extensive time and expertise from experienced artists. To solve this limitation, some methods [29], [30], [31] utilize optical flow or other techniques like facial landmarks as supervision to deform the template models. However, a large amount of hyper-parameters need to be carefully tuned toward different identities in these methods, limiting these methods' generalization. Besides, these methods are computationally slow to meet the requirements of actual use. Another category of methods [23], [24], [25] directly regresses models from images. Nevertheless, these methods rely on a large amount of training data and face difficulties when extending to other capture systems. Besides, out-of-domain expressions cannot be handled owing to insufficient training data. In this paper, we propose a novel approach for acquiring registered head models of a similar quality to those typically achieved by artists manually, but in significantly less time.

B. Facial UV-Texture Recovery

Traditional CG pipelines mainly use inverse rendering on reconstructed meshes to acquire textures from images. To accelerate the process, many methods [13], [18], [28], [46], [47], [48], [49], [50] directly extract features from images to generate textures, where the quality is limited in resolution. Despite employing super-resolution networks [18], [25], they may still arise artifacts in the textures, and cannot accurately replicate pore-level details. Moreover, the aforementioned approaches are primarily designed for static tasks, and thus cannot ensure the temporal stability of textures. Notably, Zhang et al. [28] can achieve video-level texture generation. However, it can only produce wrinkle maps that can be composited with natural high-resolution textures to represent varied expressions, rather than directly generating textures with high-frequency details, limiting its applicability. Compared to these approaches, our method ensures temporally topological consistency in texture recovery and can directly generate textures in native 8 K resolution with pore-level details.

C. 3D Scene Representation

Neural Radiance Fields (NeRF) [51] has garnered significant attention for its remarkable capability to faithfully preserve both geometric and texture details of objects. Subsequent advancements in training speed [52], [53], [54], inference speed [55], [56], [57], geometric quality [58], [59], [60], rendering quality [61], [62], [63], and dynamic scene representation [64], [65], [66], [67] have considerably enhanced the applicability of NeRF. Furthermore, 3D Gaussian splatting (3DGS) [33] has achieved

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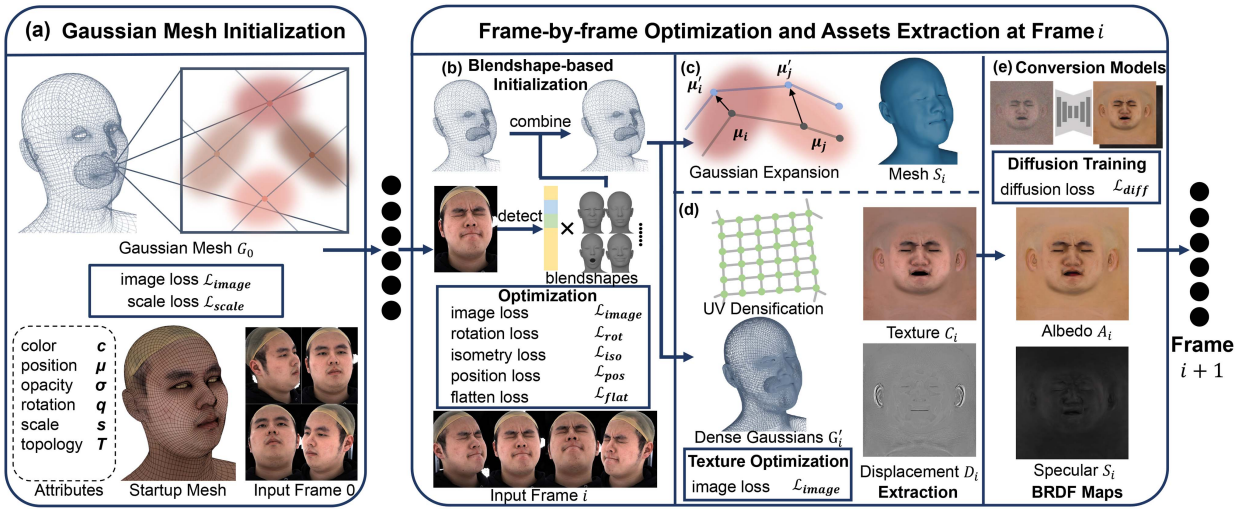


Fig. 2. Overall pipeline of our Topo4D++ framework. **(a)** We initialize Gaussian attributes and establish topological correspondence with the startup mesh. **(b)** Take one frame as an example, the Gaussian meshes are first initialized with blendshape-based expression priors. Then, geometry-related attributes in the Gaussian Mesh of the last frame are optimized by this frame under a set of topology-aware loss items. **(c)** We align the Gaussian surface with the rendering surface by Gaussian normal expansion to extract more precise meshes. **(d)** To learn pore-level ultra-high resolution texture and displacement maps, we build a dense mesh by densifying Gaussians in UV space. **(e)** Finally, we can employ our conversion model to generate albedo maps and specular maps for physically-based rendering.

SOTA results in scene representation due to its high-fidelity rendering, efficient training, and inference speed, as well as memory efficiency. This technique has been further extended to 4D scene reconstruction [34], [35], [36], [37], [38], with notable applications in representing dynamic heads [68], [69], [70] and bodies [71], [72], [73]. However, these methods typically utilize continuous neural networks or random Gaussians to represent objects, posing challenges in extracting meshes with fixed topological structures, thus limiting their integration with existing industrial CG processes. To address these issues, we propose a substantially improved 3DGS framework that can extract dynamic high-quality meshes and photo-realistic textures in constrained topology and UV structure from multi-view videos. Additionally, our generated head assets can be directly applied to the current computer graphics industrial pipelines.

III. Methods

Our method aims to achieve temporally stable head mesh reconstruction and BRDF UV-texture recovery from calibrated multi-view videos. Specifically, given sets of multi-view image sequences $\{\mathbf{I}_i^j \in \mathbb{R}^{h\times w\times 3} | 0 \le i \le F-1\}_{j=1}^{K}$ in the resolution of $h \times w$, encompassing F frames captured from K different viewpoints, our method aims to extract head meshes $\{\mathbf{S}_i := (V^i, T) | V^i \in \mathbb{R}^{n_v\times 3}\}_{i=0}^{F-1}$ in the pre-defined fixed topology T where n_v represents the number of vertices. Moreover, our method requires generating BRDF texture maps, such as Albedo maps $\{\mathbf{A}_i\}_{i=0}^{F-1}$, Specular maps $\{\mathbf{S}_i\}_{i=0}^{F-1}$ and Displacement maps $\{\mathbf{D}_i\}_{i=0}^{F-1}$ at the resolution of $8192 \times 8192 \times 3$.

To begin with, we give a brief review of 3D Gaussian Splatting [33] (Section III-A). Our method first builds a *Gaussian Mesh* by initializing a topology-integrated Gaussian set based on the facial template in the first frame (Section III-B). Then, for each subsequent frame, we alternatively perform geometry optimization and texture optimization, to learn dynamic high-fidelity geometries and textures (Section III-C). After that, we introduce the approach to extract geometries from *Gaussian Mesh* and recover textures and displacement maps (Section III-D). Finally, we propose a diffusion-based model to transfer texture maps to specular maps and albedo maps (Section III-E). The full pipeline is illustrated in Fig. 2.

A. Preliminary

3D Gaussian Splatting (3DGS) [33] is proposed as a competitive solution for photo-realistic rendering. Different from other implicit methods, 3DGS explicitly maintains a set of Gaussian distributions to model a scene. In 3DGS, each ellipsoidal Gaussian features a learnable color component $\boldsymbol{c}$ and an opacity component $\boldsymbol{\sigma}$, and is described by a covariance matrix Σ and its mean position $\boldsymbol{\mu}$:

$$G(\boldsymbol{x}) = e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}, \tag{1}$$

where Σ is further decomposed into rotation matrix R and scaling S :

$$\Sigma = RSS^TR^T. \tag{2}$$

In rendering, the color $\boldsymbol{C}$ of a pixel is acquired by sampling and blending all Gaussians that overlap the pixel in depth order:

$$\boldsymbol{C} = \sum_{i=1} \boldsymbol{c}_i\alpha_i \prod_{j=1}^{i-1}(1-\alpha_j), \tag{3}$$

where the blending weight α_i is given by evaluating a 2D Gaussian with its covariance multiplied by its opacity $\boldsymbol{\sigma}$.

However, 3DGS is designed for realistic rendering, which lacks inherent topological relationships, thus unconstrained optimization of their attributes may only yield irregular geometry. As a result, directly extracting topologically fixed meshes becomes unfeasible.

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B. Gaussian Mesh for Topology Integrated Gaussians Initialization

Extracting topologically consistent meshes from Gaussians is challenging due to the lack of geometric constraints during optimization. To this end, we propose **Gaussian Mesh**, which uniquely incorporates the topological prior into vanilla 3DGS and will not affect its high-fidelity rendering quality. We define 3D Gaussian Meshes as $G_i = \{G_{i,j}\}_{j=1}^{n_v}$ for the ith frame, where $G_{i,j}$ covers some learnable parameters, i.e., $\{\boldsymbol{\mu}_{i,j} \in \mathbb{R}^3, \boldsymbol{q}_{i,j} \in \mathbb{R}^4, \boldsymbol{s}_{i,j} \in \mathbb{R}^3, \boldsymbol{c}_{i,j} \in \mathbb{R}^3, \boldsymbol{\sigma}_{i,j} \in \mathbb{R}\}$ defined in vanilla 3DGS for the position, rotation, scaling, color, and opacity separately of the jth vertice in pre-defined topology T.

Different from 3DGS [33] using SFM-generated messy sparse points for initialization, to directly obtain pre-defined topological information, we first initialize the Gaussian Mesh with a head mesh and texture of the first frame, which is acquired by automatic MVS and NICP algorithms. Specifically, we set the mean positions $\boldsymbol{\mu}_0$ of Gaussians to the corresponding 3D coordinates of vertices in topological order. Furthermore, to better align Gaussians with the surface, we initialize the orientation $\boldsymbol{q}_0$ of each Gaussian with the normal direction of the vertices.

After initializing the shape-related attributes ($\boldsymbol{\mu}_0$ and $\mathbf{q}_0$) of Gaussians, we then optimize their rendering-related attributes ($\boldsymbol{s}_0$, $\boldsymbol{c}_0$, and $\boldsymbol{\sigma}_0$). Concretely, we initialize the scales $\boldsymbol{s}_0$ as half of the minimum distance between each Gaussian and its one-ring neighbors and set the opacity $\boldsymbol{\sigma}_0$ to 1, to avoid color blending between multiple Gaussians. To faithfully represent the color, we also initialize the color $\boldsymbol{c}_0$ of each Gaussian by sampling the corresponding pixel on the texture map according to the UV coordinate. Finally, to learn more details in dense parts, e.g., eyes and mouth, we optimize Gaussian's rotation $\boldsymbol{q}_0$ and scale $\boldsymbol{s}_0$ between the first frames $\mathbf{I}_0$ and rendered images $\mathbf{I}'_0$ with the same loss function as 3DGS:

$$\mathcal{L}_{image} = (1 - \lambda_{image})\mathcal{L}_1(\mathbf{I}_0, \mathbf{I}'_0) + \lambda_{image}\mathcal{L}_{D-SSIM}(\mathbf{I}_0, \mathbf{I}'_0). \tag{4}$$

Moreover, Gaussians have volume, leading to a certain gap between the center of Gaussians and the true surface. Therefore, the thickness of the Gaussian in the normal direction should be as small as possible. Therefore, we propose a scale loss to encourage the minimum value of every Gaussian's scale close to 0 and penalize Gaussians with a scale exceeding λ_{init} times than its initial value $\boldsymbol{s}_{init,i}$:

$$\mathcal{L}_{scale} = \sum_{i \in G} \left(\|\boldsymbol{s}_{0,i}\|_{-\infty} + max(0, \boldsymbol{s}_{0,i} - \lambda_{init}\boldsymbol{s}_{init,i}) \right). \tag{5}$$

Overall, the final loss to initialize the Gaussian Mesh can be formulated as:

$$\mathcal{L}_{init} = \mathcal{L}_{image} + \lambda_{scale}\mathcal{L}_{scale}. \tag{6}$$

C. Alternative Geometry and Texture Optimization

After initializing the Gaussian Mesh G_0, we propose an **alternative geometry and texture optimization method** to acquire the Gaussian Mesh geometry and learn varying texture colors. Specifically, at frame t, to learn accurate facial expression models, we first initialize the model G_t^0 by expression priors and the optimized model G_{t-1} in the $(t-1)$th frame. Then, we optimize the Gaussians G_t by tracking G_t^0 under the regularization of topology and physics. Afterward, dense texture color can be learned based on the tracked geometry. Generally, when a face makes wrinkles, it will appear dark since light is blocked, so static textures make it difficult to capture the geometric changes effectively. Hence, we make colors learnable in the optimization process. In addition, since the animation of the face is not a sudden process where the changes between frames are small and controllable, performing such an alternative optimization process once per frame is enough to learn dynamic textures. Subsequently, we introduce blendshape-based expression initialization, geometry optimization, and texture optimization, respectively.

Blendshape-based Expression Initialization: The human face is not a perfectly smooth surface, and during extreme expressions such as blinking or pursing the lips, the eyelids and mouth experience folds and stretching. These complex deformations are difficult to capture solely through image-based optimization. Hence, drawing from expression registration methods [74], we utilize blendshape predictions as priors to enhance the registration of extreme facial expressions. Specifically, we first create a complete set of 52 ARKit-based blendshapes $\mathbf{E} = \{\mathbf{E}_i | \mathbf{E}_i \in \mathbb{R}^{n_v \times 3}\}_{i=0}^{51}$, representing expressive morph for our topology model, which are general rather than personalized for different identities. Then, we use the offline expression predictor MediaPipe [75], to estimate the blendshape coefficients $\{\mathbf{b}^t | \mathbf{b}^t \in \mathbb{R}^{52}\}_{t=0}^{F-1}$ for the entire video sequence, followed by linearly smoothing to remove any outlier values. Finally, the predicted values are multiplied by the expression coefficients and added to the optimization result of the $(t-1)$th frame, serving as the initial values G_t^0 for the tth frame as:

$$G_t^0 = G_{t-1} + (\mathbf{b}^t - \mathbf{b}^{t-1}) \times \mathbf{E}. \tag{7}$$

Subsequently, the initialized results are used for the geometric optimization of the ith frame. It is worth noting that facial expressions vary significantly among individuals, thus further optimization is required to capture the personalized expressions of each identity accurately.

Geometry Optimization: Naively optimizing the Gaussian Mesh causes topological confusion. To maintain the topology and regular mesh arrangement within Gaussians, we extend 3DGS [33] by introducing physical and topological priors. Specifically, we propose a physical loss item $\mathcal{L}_{phy}$ that constrains local rigidity and topological loss $\mathcal{L}_{topo}$ to improve the mesh quality. Overall, the loss functions for geometry optimization are constructed in three parts:

$$\mathcal{L}_{geo} = \mathcal{L}_{image} + \mathcal{L}_{phy} + \mathcal{L}_{topo}. \tag{8}$$

Physical Prior Loss: Solely utilizing color as supervision to optimize Gaussians is problematic since low-frequency texture details, e.g., forehead and cheek, lead to point mistracking, thus breaking the facial topology. To better regularize the motion of Gaussians and maintain topological information, we modify the loss functions in Luiten et al. [34] with the constraints by

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one-ring neighbors as:

$$\mathcal{L}_{rot} = \frac{1}{2n_e} \sum_{i \in G} \sum_{j \in \mathcal{K}_i} w_{i,j} \|\hat{\boldsymbol{q}}_{t,j} \hat{\boldsymbol{q}}_{t-1,j}^{-1} - \hat{\boldsymbol{q}}_{t,i} \hat{\boldsymbol{q}}_{t-1,i}^{-1}\|_2, \quad (9)$$

where $\hat{\boldsymbol{q}}$ is the normalized quaternion, n_e is the number of edges, and $\mathcal{K}_i$ means the one-ring neighbours of $G_{t,i}$. The loss weighing factor $w_{i,j}$ takes into account the edge length of the Gaussian Mesh at the first frame:

$$w_{i,j} = exp(-\lambda_w \|\boldsymbol{\mu}_{0,j} - \boldsymbol{\mu}_{0,i}\|_2^2). \quad (10)$$

In addition to the rotation similarity, we find that long-term physical loss is important for maintaining long-term stable dense correspondence:

$$\mathcal{L}_{iso} = \frac{1}{2n_e} \sum_{i \in G} \sum_{j \in \mathcal{K}_i} w_{i,j} \big| \|\boldsymbol{\mu}_{0,j} - \boldsymbol{\mu}_{0,i}\|_2 - \|\boldsymbol{\mu}_{t,j} - \boldsymbol{\mu}_{t,i}\|_2 \big|. \quad (11)$$

Finally, our physical loss items are the weighted sum of these two loss functions:

$$\mathcal{L}_{phy} = \lambda_{rot} \mathcal{L}_{rot} + \lambda_{iso} \mathcal{L}_{iso}. \quad (12)$$

Topological Prior Loss: Physical constraints achieve long-term tracking of the corresponding Gaussians, but they will lead to irregular wiring and unsmooth surfaces. To realize regular wiring, we calculate the L2 loss between the position of each vertex and the average position of its neighbors:

$$\mathcal{L}_{pos} = \frac{1}{n_v} \sum_{i \in G} \left(\boldsymbol{\mu}_{t,i} - \frac{\sum_{j \in \mathcal{K}_i} \boldsymbol{\mu}_{t,j}}{|\mathcal{K}_i|} \right)^2. \quad (13)$$

To maintain the stable topology and smooth surface, we further apply mesh flattening loss to the angles between adjacent faces in optimization:

$$\mathcal{L}_{flat} = \sum_{\theta_i \in e_i} \left(1 - cos(\theta_{t,i} - \theta_{0,i})\right), \quad (14)$$

where $\theta_{t,i}$ is the angle between the faces on the mesh that have the common edge e_i at frame t. Overall, our topological prior loss servers as:

$$\mathcal{L}_{topo} = \lambda_{pos} \mathcal{L}_{pos} + \lambda_{flat} \mathcal{L}_{flat}. \quad (15)$$

Texture Optimization: After completing the geometry optimization, we then fix the location of each vertex and optimize the color of each Gaussian with (6). This process produces an initial rough color estimation for the optimization in subsequent frames. Note that, due to the limited number of Gaussians, the aforementioned optimization process is performed on bilinearly interpolated low-resolution images.

However, coarse geometry and textures lack sufficient details for high-fidelity rendering. We continue to capture high-detailed texture and geometry with pore-level details on high-resolution images. Different from vanilla 3DGS, which adopts an adaptive densification process that results in a messy topology, we propose a novel densification method, i.e., **UV Space Densification**, which allows for topologically densified Gaussians corresponding to the UV space of the texture map. Specifically, at the first frame, we initialize the dense Gaussian Mesh G_0' with the Gaussians in G_0. When performing UV Space Densification, we insert each quadrilateral grid in G_0' with $(N \times N)$ smaller equidistant grids. Specifically, the newly generated Gaussian's position, UV coordinates, and color are obtained by bilinear interpolation sampling the corresponding attributes of the four Gaussians at the original grid vertices:

$$A_{i,j} = \frac{1}{(N-1) \times (N-1)} \begin{bmatrix} N-1-i & i \end{bmatrix} \begin{bmatrix} A_{0,0} & A_{0,N-1} \\ A_{N-1,0} & A_{N-1,N-1} \end{bmatrix} \begin{bmatrix} N-1-j \\ j \end{bmatrix}, \quad (16)$$

where A can be color, UV coordinate, and position. Most importantly, we can easily establish topological relationships between Gaussians, which is equivalent to subdividing the grid and inserting more sampling points in UV space.

We optimize dense Gaussian Mesh G_t' by (4) and follow the same practice as Section III-B to obtain high-frequency texture and geometry with sub-micron details. To obtain geometric details, we optimize displacement maps by constraining the densified Gaussian positions, which move only along their normal directions. And for the texture, we do not impose any additional constraints. In implementation, we only perform UV densification once in the first frame and compute every Gaussian's interpolation weights given by (16). In the texture optimization stage of each subsequent frame, we use these weights to calculate the attributes of each Gaussian in G_t' for dense Gaussian Mesh initialization.

D. Extracting Geometry and Texture From Gaussians

After completing the geometry and texture optimization, we can extract 4D meshes from $\{G_i\}_{i=0}^{F-1}$, 8K texture maps and displacement maps from $\{G_i'\}_{i=0}^{F-1}$, respectively.

Geometry Extraction: Despite special regularization for the scale of the Gaussians, it still has volume, resulting in a slight decrease in the geometry reconstructed by directly extracting the Gaussians' positions. Therefore, we propose Gaussian Normal Expansion to make surfaces derived from Gaussian surfaces resemble real surfaces more closely. As illustrated in Fig. 2(c), we offset each Gaussian in the direction of the vertex normal by its projection scale in the normal direction to obtain the final meshes. Specifically, we offset each Gaussian's mean position $\boldsymbol{\mu}$ by its projection distance from the surface of the ellipsoid in the vertex normal direction:

$$\boldsymbol{n}' = R^{-1} \boldsymbol{n}, \quad (17)$$

$$\boldsymbol{\mu}' = \boldsymbol{\mu} + \sqrt{\frac{1}{\frac{n_x'^2}{s_x^2} + \frac{n_y'^2}{s_y^2} + \frac{n_z'^2}{s_z^2}}} \boldsymbol{n}, \quad (18)$$

where $\boldsymbol{n}$ is the vertex normal of each Gaussian.

Texture Extraction: To generate 8 K texture and displacement maps from the dense Gaussian meshes $\{G_i'\}_{i=0}^{F-1}$, we map Gaussians' colors and geometric offsets to UV space based on their UV coordinates. Since Gaussian meshes are topologized, we can triangulate them and map learned dense attributes to texture maps $\{\mathbf{C}_i\}_{i=0}^{F-1}$ or displacement maps $\{\mathbf{D}_i\}_{i=0}^{F-1}$ by a

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rasterization-based rendering method [76]. Specifically, if the Gaussian indices of the visible triangles for a pixel at (x, y) are denoted as $i_{0,1,2}$ and the center of mass is denoted as $w_{0,1,2}$, we can calculate the color $\boldsymbol{C}_{x,y}$ and geometric offset $\boldsymbol{D}_{x,y}$ of the pixel at (x, y) as:

$$\boldsymbol{C}_{x,y} = w_0 \boldsymbol{c}_{i_0} + w_1 \boldsymbol{c}_{i_1} + (1 - w_0 - w_1)\boldsymbol{c}_{i_2}, \tag{19}$$

$$\boldsymbol{D}_{x,y} = w_0 \boldsymbol{d}_{i_0} + w_1 \boldsymbol{d}_{i_1} + (1 - w_0 - w_1)\boldsymbol{d}_{i_2}. \tag{20}$$

E. Conversion Models for BRDF Assets Generation

To enable rendering facial animations under different lighting conditions, BRDF-based texture maps are necessary. However, the obtained texture maps are coupled with ambient lighting and fail to capture specular highlights. Therefore, we aim to train a conversion network to generate consistent albedo maps $\boldsymbol{A}$ and specular maps $\boldsymbol{S}$ from the captured texture maps $\boldsymbol{C}$.

We first construct a private multi-view BRDF dataset consisting of 100 different identities with about 30 expressions. The dataset is captured in a Light Stage system [77], where each subject is illuminated by 15 different polarized lighting conditions within 2 seconds, and recorded simultaneously from 7 static camera viewpoints. We then compute the albedo and specular components for each viewpoint by photometric techniques [3]. Notably, among the 15 lighting conditions, one corresponds to a constant uniform illumination that matches the dynamic capture setup, benefiting the accurate texture conversion. The camera system is color-calibrated to ensure the consistency between images captured by dynamic and static cameras, which is crucial for accurate training of subsequent BRDF conversion models.

Given that the system's reference illumination remains fixed, lighting variation does not need to be explicitly modeled. Consequently, the captured textures exhibit a stable mapping to albedo and specular components. Therefore, our goal is to use a single model to disentangle the albedo and specular maps directly from captured textures. However, two challenges arise: **1)** the limited dataset size makes it difficult to directly learn the correlations across different channels at the image level, and **2)** the learned features may not generalize well to unseen identities. To address these issues, we leverage the large-scale pretrained models to extract well-embedded features. Specifically, we employ the frozen VAE's encoder $\mathcal{E}$ and decoder $\mathcal{D}$ from Stable Diffusion [78] to encode different maps. Since this VAE is pretrained on massive datasets, it is capable of robustly encoding albedo, specular, and texture images into a latent space and enabling high-quality bi-directional transformation between high-dimensional images and compact latent codes. This provides two major benefits: strong generalization and computational efficiency. Thus, the latent representations $\{\mathbf{z}^{\mathbf{C}}, \mathbf{z}^{\mathbf{A}}, \mathbf{z}^{\mathbf{S}}\}$ of $\{\boldsymbol{C}, \boldsymbol{A}, \boldsymbol{S}\}$ can be obtained by:

$$\mathbf{z}^{\mathbf{C}}, \mathbf{z}^{\mathbf{A}}, \mathbf{z}^{\mathbf{S}} = \mathcal{E}(\mathbf{C}, \mathbf{A}, \mathbf{S}). \tag{21}$$

Following, our objective is to train a diffusion model to convert textures to albedo and specular maps. We adopt the model architecture of Stable Diffusion [78] for training, but without introducing external guidance. Instead, we directly use the texture latents as part of input to provide the condition. In the training process, the random Gaussian noise $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is added to the concatenated latents $\mathbf{z}^{\{\mathbf{A},\mathbf{S}\}}$ at the step t to get the final noisy target $\mathbf{z}_T^{\{\mathbf{A},\mathbf{S}\}}$:

$$q\left(\mathbf{z}_{1:T}^{\{\mathbf{A},\mathbf{S}\}} \mid \mathbf{z}^{\{\mathbf{A},\mathbf{S}\}}\right) = \prod_{t=1}^{T} q\left(\mathbf{z}_t^{\{\mathbf{A},\mathbf{S}\}} \mid \mathbf{z}_{t-1}^{\{\mathbf{A},\mathbf{S}\}}\right), \tag{22}$$

where $\mathbf{z}_t^{\{\mathbf{A},\mathbf{S}\}} = \sqrt{1-\beta_t}\,\mathbf{z}_{t-1}^{\{\mathbf{A},\mathbf{S}\}} + \sqrt{\beta_t}\,\varepsilon_t$, with β_t to control the quality of noise.

Later, the texture latents $\mathbf{z}^{\mathbf{C}}$ and noisy latents $\mathbf{z}_T^{\{\mathbf{A},\mathbf{S}\}}$ are fed into the diffusion model θ to predict the noise $\epsilon_\theta(\mathbf{z}_t^{\{\mathbf{A},\mathbf{S}\}}; t)$ by the reverse Markov process. Attention models are adopted between the albedo and specular maps to explore their connection. The optimization target is to minimize the predicted noise and the randomly sampled noise $\boldsymbol{\epsilon}_t$ in every step t by:

$$\mathcal{L}_{\text{Diff}} = \mathbb{E}_{t,\epsilon}\left[w(t)\left\|\epsilon_\theta\left(\mathbf{z}_t^{\{\mathbf{A},\mathbf{S}\}}; t\right) - \epsilon_t\right\|_2^2\right], \tag{23}$$

where $w(t)$ is a weighting function.

In the inference process, the texture latents $\mathbf{z}^{\mathbf{C}}$ and the sampled random noise ϵ are concatenated and passed through the diffusion model θ to obtain the final albedo latent $\{\mathbf{z}^{\mathbf{A}'}, \mathbf{z}^{\mathbf{S}'}\}$. Finally, the corresponding albedo A' and specular S' maps can be recovered via the decoder $\mathcal{D}$ of Stable Diffusion as:

$$\mathbf{A}', \mathbf{S}' = \mathcal{D}(\mathbf{z}^{\mathbf{A}'}, \mathbf{z}^{\mathbf{S}'}). \tag{24}$$

It is worth noting that $\mathcal{E}$ and $\mathcal{D}$ are designed to handle images with a maximum resolution of 768×512. Therefore, we randomly crop high-resolution images to learn local correspondences during training. In the inference process, high-resolution texture maps are cropped sequentially, processed individually, and then concatenated to obtain 8K albedo maps and specular maps.

IV. Benchmark: JHead

To fill the gap that lacks a comprehensive and effective evaluation for 4D head capture, we propose a novel benchmark, JHead. We first utilize our Light Stage system to capture multi-view face performance. As shown in Fig. 3(a), our Light Stage system consists of 24 time-consistent dynamic cameras capable of capturing multi-view videos at 60FPS with a resolution of 4096×3000 pixels. Each camera is hardware-controlled with a time error of less than 1 microsecond. The 24 cameras are precisely calibrated to obtain accurate intrinsic and extrinsic parameters. To minimize perspective errors caused by facial shadows, we employ multiple surrounding uniform light sources. Subjects are constrained to chairs during performances, maintaining relatively static head movements, which aligns with the facial capture requirements in industrial production processes.

During filming, we capture videos for 20 identities with 1 segment of video each, lasting approximately 20 seconds. In 10 segments, the subjects spontaneously recite a lengthy passage to simulate natural speech conditions. In the other 10 segments, the subjects randomly change expressions, especially extreme expressions, involving severe facial distortions and wrinkles, to validate the method's generality and adaptability to extreme

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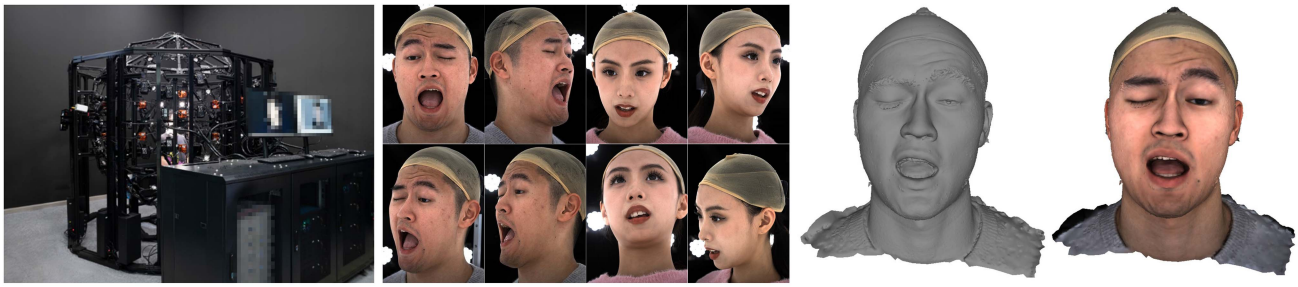


Fig. 3. We show (a) the diagram of our Light Stage capture system, (b) some raw data of different identities, and (c) high-quality head scans.

TABLE I
THE INITIALIZATION SETTINGS AND LEARNING STRATEGIES FOR EACH GAUSSIAN ATTRIBUTE

Symbol	First Frame Opt.	Geometry Opt.	Texture Opt.	Learning Rate
$\boldsymbol{\mu}$	Fixed	Learnable	Along with Normal	$1.6e^{-5}$
$\boldsymbol{q}$	Learnable	Learnable	Fixed	$1e^{-3}$
s	Learnable	Fixed	Fixed	$1e^{-3}$
σ	Fixed	Fixed	Fixed	0.0
$\boldsymbol{c}$	Fixed	Fixed	Learnable	$2.5e^{-3}$

scenarios. Each sequence begins from **a neutral expression**. We show some examples in Fig. 3(b). We have obtained permission from the volunteers to use the collected data for research and public release.

After capturing multi-view videos, we perform MVS reconstruction to build head scans for each frame using Agisoft Metashape software [39], as shown in Fig. 3(c). These head models contain about 200000 vertices, which can keep most details on the faces. To evaluate the effectiveness of head model registration, the point-to-surface distance can be calculated between the scans and the face model generated by these approaches.

V. EXPERIMENTS

A. Implementation Details

Head Reconstruction: As shown in the Table I, we adopt different optimizing strategies at different stages in our pipeline. **1)** When optimizing the initial Gaussian Mesh at the first frame, we only optimize $\boldsymbol{q}$ and $\boldsymbol{s}$, and keep others fixed. **2)** During geometry optimization, we optimize these geometry-related attributes ($\boldsymbol{\mu}$ and $\boldsymbol{q}$) to track Gaussian motions. **3)** During texture optimization, we initialize a dense Gaussian Mesh by sampling each Gaussian's position by (16) and set their opacity to 1. We also fix the Gaussian scale to be the minimum distance from their one-ring neighbors to prevent color overlapping. Since our Light Stage can provide uniform lighting, we use RGB rather than SH in 3DGS to represent view-consistent colors, which is more efficient and easier to optimize. In our setting, we employ a mesh topology consisting of $n_v = 8280$ vertices and $n_f = 16494$ faces, where Gaussians are placed at the center of each vertex, resulting in a total of 8280 Gaussians. We implement our method based on PyTorch [79] and an NVIDIA 3090 GPU, optimized by Adam [80]. We employ an MVS method [81] to reconstruct 4D head scans for evaluation and use an automatic iterative closest point (ICP) [4] algorithm to obtain a roughly accurate mesh in the first frame for initialization. The geometry optimization stage includes 1000 iterations at each timestamp, with 16 images resized to 512×375. The expression prior is initialized using ARKit-compatible blendshapes. Specifically, we first construct various 52 expression meshes based on the template, and compute the difference between each expression and the template to form the blendshapes. It is important to note that these blendshapes are generalized and not tied to any specific identity. In the subsequent optimization stage, we first estimate the value of blendshape from a frontal view using MediaPipe [75] and then apply the blendshapes to the model. Although MediaPipe only provides coarse estimates, the following optimization process refines them, resulting in sufficiently accurate dynamic facial meshes. We mask out the inner mouth using a face parsing model [82] to prevent the vertices around the lips from learning incorrect colors. The texture optimization stage consists of 300 iterations at each timestamp and is learned at the original 16 4K-resolution images without preprocessing, with a dense number $N = 30$. The texture outside the facial area is obtained by $\alpha-$blending with the template textures. We set $\lambda_{image} = 0.2$, $\lambda_{scale} = 10$, $\lambda_{rot} = 20$, $\lambda_{iso} = 20$, $\lambda_{pos} = 1e3$, and $\lambda_{flat} = 2e - 4$. All hyper-parameters remain the **same** for all experiments and different datasets.

BRDF Generation: We employ the Stable Diffusion V2 model [78], pre-trained on the LAION-5B dataset [83], as the base model. In the training of diffusion-based conversion models, the VAE is frozen to extract generalized features. The captured image and calculated BRDF images are randomly cropped into a resolution of 512×512, and fed into the diffusion model. The model is also optimized with the Adam optimizer, and the learning rate is 1e-5, trained on Pytorch and 4 Nvidia A6000 GPUs, with a batch size of 128.

B. Evaluation of Face Reconstruction

We evaluate our method on our JHead Benchmark and compare it to 2 types of SOTA topology-consistent face reconstruction methods: (1) single-view methods: DECA [7], 3DDFA _ V2 [21], and HRN [49]; (2) multi-view methods: MVFR [26], DFNRMVS [12], and VHAP [22]; and some NICP-based methods: Amberg et al. [84] and perfessional software Wrap4D [5], both of which are constrained with facial landmarks. It is worth

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TABLE II
QUANTITATIVE EVALUATION FOR DIFFERENT FACE RECONSTRUCTION METHODS ON OUR JHEAD BENCHMARK. WE MEASURE THE PERCENTAGE OF VERTICES WITHIN DIFFERENT ERROR LEVELS AND CALCULATE THE MEAN ERROR AND MEDIAN ERROR AS METRICS.

Type	Methods	<0.2mm(%)↑	<0.5mm(%)↑	<1mm(%)↑	<2mm(%)↑	<3mm(%)↑	Mean(mm)↓	Med.(mm)↓
Single-view	DECA [7]	2.264	5.656	11.158	21.489	30.742	8.294	5.685
	3DDFA_V2 [21]	3.072	7.658	15.260	29.796	43.070	4.211	3.712
	HRN [49]	2.220	5.518	10.904	21.171	30.740	3.438	3.323
Multi-view	MVFR [26]	3.800	9.415	18.212	32.144	41.082	8.720	4.897
	DFNRMVS [12]	3.615	9.018	17.912	34.684	49.462	3.713	3.262
	VHAP [22]	9.013	22.229	42.512	72.663	88.275	1.530	1.247
	Topo4D (Ours) [41]	22.889	51.999	77.557	91.239	95.001	0.873	0.489
	Topo4D++ (Ours)	**23.704**	**53.970**	**79.032**	**91.653**	**95.200**	**0.840**	**0.458**

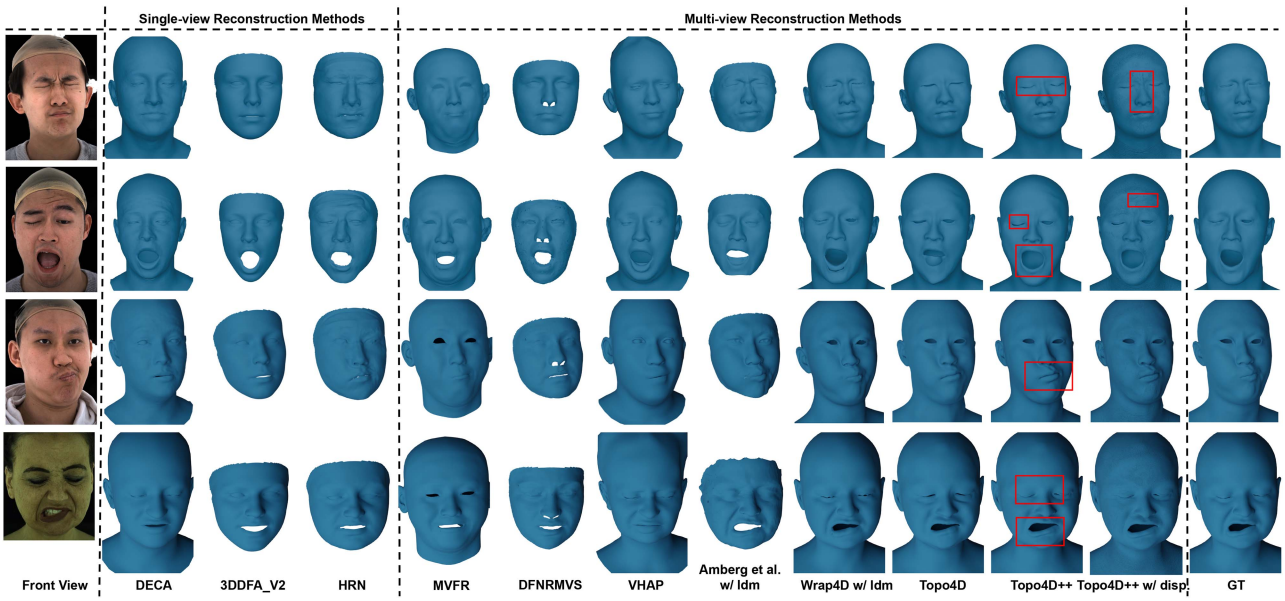


Fig. 4. Qualitative evaluation of meshes generated by our method and other topology-consistent reconstruction methods on JHead and Multiface datasets [40]. We use artist-manually registered head meshes as the ground truths. We highlight areas which are difficult to reconstruct.

noting that NICP-based methods rely on MVS to obtain high-quality scans, after which facial templates are warped to the scans by optimizing distance metrics for head registration. While these methods can achieve superior quantitative results in terms of distance error, they neglect semantic correspondence, often leading to visually incorrect results with noticeable misalignments. Even with constraints, such as facial landmarks, performance is limited in extreme expressions. Therefore, we limit our comparisons with these NICP-based methods qualitatively. For single-view methods, we input the front-view images. For multi-view methods, we use 16 views, except that DFNRMVS uses a front view and an oblique side view image to produce its best results. Moreover, to demonstrate the generalization of our Topo4D++, we also evaluate our Topo4D++ on the Multiface dataset [40]. Furthermore, we also compare our methods with some regression-based methods Tempeh [24] and ReFA [25] qualitatively to show our superior results and generalization to these methods.

Quantitative Comparisons: We evaluate facial reconstruction accuracy by mesh-to-scan distances. To minimize topology inconsistency among different approaches, we only compute metrics on the facial regions, excluding the ears, neck, and the back of the head. The results of regression methods are rigidly aligned with the scans for evaluation. As shown in Table II, our Topo4D++ significantly outperforms other methods across all metrics, with the majority of vertices falling within high-precision ranges and 53.97% more vertices achieving sub-0.5 mm accuracy. This improvement stems from the use of reliable initialization priors and rendering-based optimization, which jointly ensure accurate vertex tracking throughout the animation and yield superior registration results. Moreover, Topo4D++ surpasses Topo4D by leveraging expression priors, enabling more robust initialization under extreme expressions.

Qualitative Comparisons: Fig. 4 shows visual mesh reconstruction comparisons between different methods, where the head models registered manually by experienced artists are regarded as the ground truths. Single-view methods [7], [21], [49] suffer from detail loss due to the lack of sufficient multi-view information. Although several multi-view methods [12], [22], [26] leverage optimization across multiple views, they still fail to faithfully reconstruct fine-grained facial details. NICP-based methods [5], [84] can produce visually detailed results. However, the advanced landmark detectors still produce significant errors under some extreme expressions, resulting in semantically

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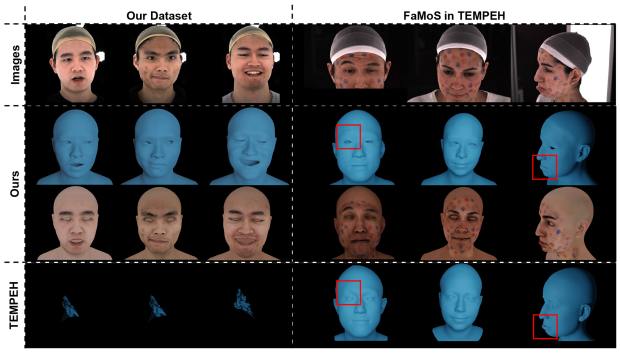


Fig. 5. Qualitative comparisons with TEMPEH [24] on our JHead benchmark and FaMoS dataset, respectively.

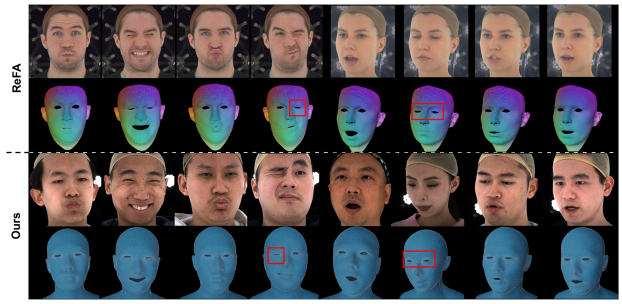


Fig. 6. Qualitative comparisons of geometries with ReFA [25]. We show examples of our dataset similar to those expressions in ReFA's paper.

incorrect registration, which introduces noticeable artifacts in highly dynamic regions such as the eyes and mouth corners. In contrast, owing to the dense correspondence, our proposed Topo4D and Topo4D++ can faithfully capture both asymmetric extreme expressions and minor facial changes, outperforming other methods. Compared to vanilla Topo4D, Topo4D++ can better identify some skin contraction, e.g., eye blink, and mouth struggle, leading to better results. In addition, thanks to the optimized displacement maps, the meshes are enriched with fine-scale details such as wrinkles and pores, further enhancing the overall visual realism.

To evaluate the robustness and generalization of our Topo4D++, we make comparisons on the recent popular datasets, e.g., Multiface [40], as shown in Fig. 4. We calibrated the Multiface dataset with the COLMAP method [85], which cannot estimate absolutely accurate camera parameters due to limited views. Notably, Topo4D++ uses the **same hyper-parameters** on MultiFace datasets as it does on JHead. The results shown in Fig. 4 show that our Topo4D++ can still perform much better than other methods, demonstrating the generalization of our method and robustness towards calibration errors.

Comparisons with TEMPEH: TEMPEH [24], trained with a large amount of 4D head data, can directly regress dynamic topological head models from multi-view images, which targets a similar task as our work. However, 2 major differences exist between TEMPEH and our Topo4D++: **1)** TEMPEH is trained on its proposed dataset FaMoS, and it cannot apply to other capture systems. Conversely, our Topo4D++ is suitable for any dataset, including FaMoS. **2)** TEMPEH can only generate meshes while our Topo4D++ can generate both meshes and high-quality textures. For fair comparisons, we compare TEMPEH and our Topo4D++ both on our JHead and TEMPEH's FaMoS dataset. Note that, due to the lack of a large amount of registered data in our capture system for supervision, we are unable to train TEMPEH by ourselves. Therefore, we use its publicly released pre-trained models for face reconstruction. As shown in Fig. 5, **1)** on our JHead, our method faithfully reconstructs facial geometry and generates 8 K textures that can produce realistic rendering results. However, TEMPEH fails to reconstruct meaningful heads, revealing that the pre-trained TEMPEH cannot be directly applied to different capture systems. **2)** FaMoS consists of 16 Gy-scale and 8 color images in each frame, including 2 color images from the back views. In the experiments, we only use 6 color images containing the front face as the input of our method, while showing TEMPEH's best results with its publicly released pre-trained model on 16 Gy-scale views. Even though, we achieve competitive geometric results with TEMPEH. Especially under some extreme conditions in the 4th and 6th columns in Fig. 5, we can even better restore asymmetrical eyebrow and pouting expressions. Additionally, our method can generate high-fidelity textures while TEMPEH cannot. **Overall**, our method is more general across capture systems than TEMPEH [24], and can generate high-quality texture maps. Moreover, our method does not require a large amount of registered data for training.

Comparisons with ReFA: Different from our Topo4D++, ReFA [25] employs a regression-based module to generate facial meshes from multi-view images, and the 4 K textures are generated via super-resolution modules rather than inverse-rendering. Since ReFA does not release its codes, pre-trained models, and test datasets, we can only borrow some examples from its manuscript and website for qualitative comparisons with our Topo4D++ toward the quality of meshes. In the comparisons, we select some expressions similar to ReFA's results for a relatively fair evaluation of extreme expressions. As shown in Fig. 6, our method can more faithfully reconstruct the detailed geometric meshes than ReFA, especially in the eyes regions, e.g., 4th and 6th columns, where ReFA cannot completely close the eyes.

C. Evaluation of Texture Generation

We qualitatively compare our generated textures with state-of-the-art facial texture estimation models: UnsupTex [48] and HRN [49] on the JHead dataset and the Multiface [40] Dataset. Fig. 7 shows the rendering results of different methods, including the generated 8 K textures. In the comparisons, UnsupTex and HRN are limited to producing low-resolution textures that lack realistic details. In contrast, our method directly generates high-quality 8 K textures without up-sampling, faithfully capturing facial wrinkles, hair, and pores, and achieving noticeably superior rendering quality. Besides, HRN can also generate displacement maps, while lacking wrinkle-level details. In contrast, our Topo4D++ can generate pore-level detailed displacements, demonstrating the effectiveness of our method.

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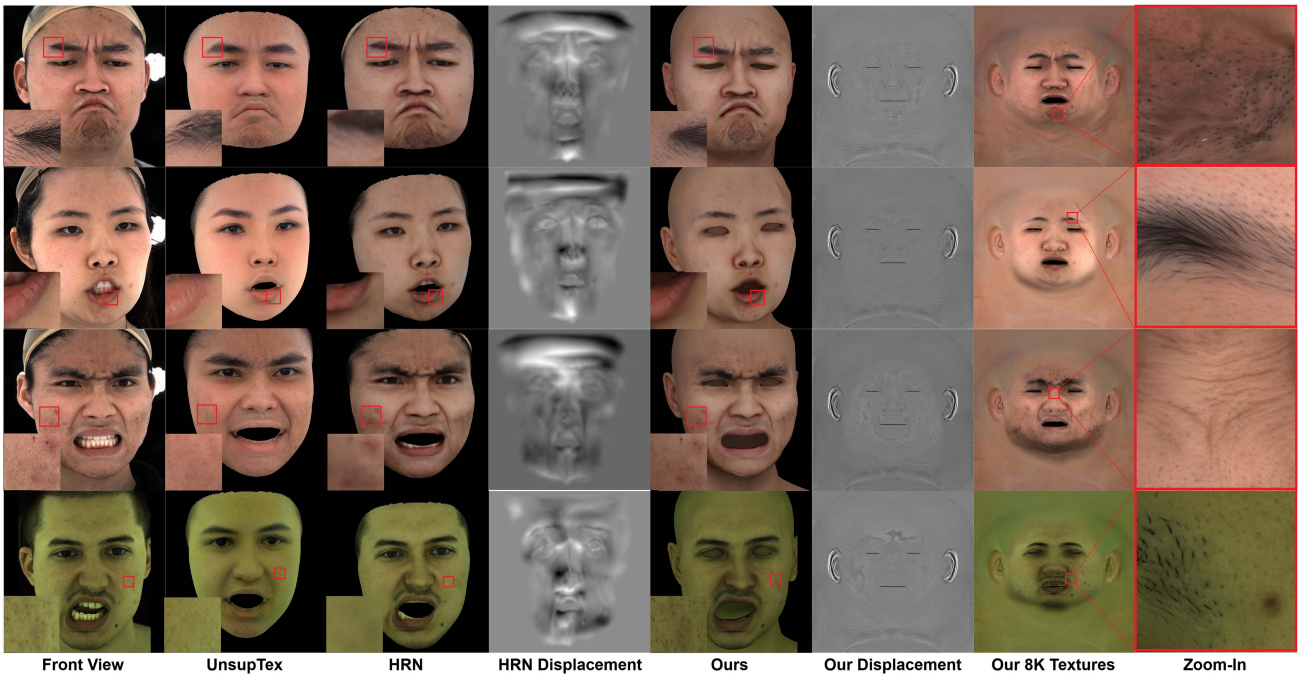


Fig. 7. Qualitative evaluation of the rendering results and displacement maps among our method, UnsupTex [48], and HRN [49] on JHead and Multiface datasets [40]. The generated 8 K textures and pore-level details are demonstrated in columns 7 and 8.



Fig. 8. Qualitative comparison of textures with ReFA [25]. We show examples of textures generated on our JHead benchmark. Please **zoom-in** for detailed observation.

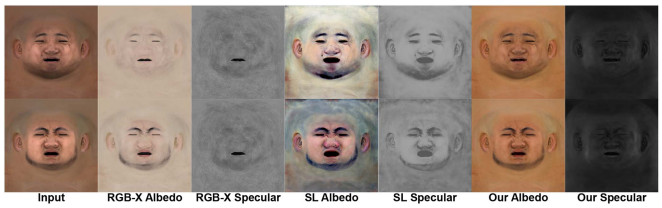


Fig. 9. Qualitative evaluation of the generated BRDF results of our method, SwitchLight [86] (short as SL) and RGB-X [87].

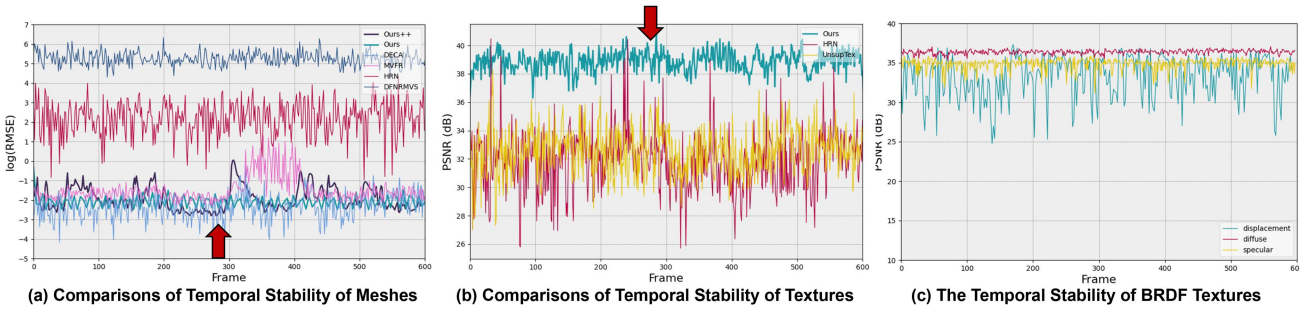
Comparisons with ReFA: As shown in Fig. 8, our Topo4D++ can faithfully generate 8 K high-resolution dynamic UV textures. Instead, ReFA [25] relies on a super-resolution module to obtain 4 K textures, so it fails to faithfully restore the original details of the face and leads to artifacts in the textures. Overall, our Topo4D++ achieves better results than RefA in texture quality.

Evaluation of BRDF texture Generation: To better demonstrate the advantages of our method in generating BRDF textures, we compare our method against recent state-of-the-art methods, RGB-X [87] and SwitchLight [86]. The results of SwitchLight are obtained with their SwitchLight Studio. Note that, we do not compare our method with AvatarMe++ [19] since it does not release its code and pre-trained models. Since these methods differ in data definitions, a fair quantitative comparison is not feasible. Therefore, we only conduct visually qualitative comparisons. As shown in Fig. 9, compared to RGB-X and Switchlight, which introduce over-smooth results and obvious artifacts, our method better preserves identity attributes in the texture and exhibits finer details, such as wrinkles, achieving superior performance in BRDF texture synthesis.

D. Evaluation of Efficiency

To meet the requirements of practical 4D head reconstruction, efficiency is essential. We first report the image resolution processed at each step, together with the corresponding averaged runtime and GPU memory usage in Table III, which are evaluated with 100 frames on a single NVIDIA RTX 3090 GPU. Since relatively accurate geometry is required, initializing and optimizing the Gaussian mesh for the first frame takes much time. However, compared to reconstructing hundreds or thousands of frames in a dynamic performance, this overhead is negligible. In the subsequent processing stages, accurate geometry, texture maps, and displacement maps can be obtained within approximately 90 seconds. Considering the large **8 K** resolution of images, computing the BRDF maps requires around an additional 2.5 minutes, which is acceptable

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(a) Comparisons of Temporal Stability of Meshes (b) Comparisons of Temporal Stability of Textures (c) The Temporal Stability of BRDF Textures

Fig. 10. Comparisons of temporal stabilization on our JHead dataset, and our method is indicated by red arrows. **(a)** The curves of log(RMSE) (lower is better) of several face reconstruction methods. **(b)** The curves of PSNR (higher is better) are calculated between textures of adjacent frames. **(c)** The curves of PSNR are calculated between BRDF maps of adjacent frames to demonstrate the stability of the conversion model.

TABLE III
THE ILLUSTRATION OF THE TIME AND GPU MEMORY USAGE IN EACH STEP

Stage	Step	Res.	Runtime (s)	GPU Memory (GB)
1st Frame	Mesh Acqu.	4K	≈ 240	3
	Gaussian Mesh Init.	4K	≈ 180	0.37
	UV Space Dens.	-	≈ 55	0.54
	Total	4K	475	0.54
Following Frames	Blendshape	512	<0.01	<0.01
	Mesh Opt.	512	30.71	0.49
	Dis. Opt.	4K	27.34	9.28
	Tex. Opt.	4K	28.25	9.28
	BRDF Acqu.	8K	142.63	8.47
	Total	-	228.93	9.28

TABLE IV
EFFICIENCY COMPARISONS AMONG OPTIMIZATION-BASED METHODS PER FRAME. NOTE THAT THE RESULTS DO NOT CONTAIN THE BRDF TEXTURE ACQUISITION

Methods	Res.	Runtime (s)	GPU Memory (GB)	Hand-Craft
MVS+ICP w ldm	4K	≈ 240	≈ 3	✗
VHAP	512	40	6.50	✗
MVS+Wrap4D	4K	≈ 600	≈ 3	✓
Topo4D++	4K	86.30	9.28	✗

in practice. The total runtime is about 229 seconds per frame. Regarding GPU memory usage, since the optimization process and BRDF computation can be performed separately, the peak memory consumption is only 9.28 GB, which can be supported by most consumer-grade GPUs.

In addition, we compare the efficiency of our proposed method with recent optimization-based approaches in Table IV. It is worth noting that regression-based methods are indeed highly efficient, while their accuracy is far from sufficient to meet the requirements of high-precision reconstruction. Compared with the MVS+NICP pipeline, our method achieves 3 times higher efficiency with more accurate registered models. Relative to VHAP, which obtains geometry and texture at the 512 resolution, our method operates at 4 K resolution images with only slightly higher runtime and memory usage, while delivering substantially better results. Furthermore, compared with current industrial software such as Wrap4D [5], which requires manual intervention by highly experienced artists, our method is far more efficient and can achieve competitive reconstruction results. Overall, our approach achieves an optimal balance of quality and efficiency, making it well-suited for practical asset production.

E. Temporal Stability Comparisons

We compare the temporal stability of the generated meshes and textures with other methods. For meshes, we measure geometry consistency by calculating the RMSE between adjacent frames. Fig. 10(a) shows the curve of each method over an expression sequence. Our method achieves much more temporally stable vertex tracking with extreme expressions, whereas other methods exhibit considerable performance fluctuations. We notice that, in some cases, the inter-frame difference of DECA [7] is lower than ours. This is because DECA fails to capture some extreme expressions or minor facial changes, as shown in Fig. 4, and therefore tends to keep the meshes unchanged. Instead, our method can faithfully capture facial motions while maintaining temporal consistency. From the perspective of textures, we measure stability by comparing the PSNR of textures between adjacent frames. As shown in Fig. 10(b), our method has the highest PSNR and lowest variance, indicating superior temporal stability. Moreover, our generated BRDF textures can also remain temporally stable (Fig. 10(c)). In conclusion, our method demonstrates better temporal consistency compared to other methods in both geometry and textures.

F. Ablation Studies

In this section, we evaluate the impacts of the crucial designs and parameters used in our method. For geometry reconstruction, we ablate loss items, the expression prior, the number of input views, and the Gaussian normal expansion operation, as shown in Fig. 11 and Table V, which are implemented with 16 views, except for the view ablation. Regarding texture estimation, we explore the impact of UV densification density on texture quality, along with the influence of $\mathcal{L}_{scale}$ in Fig. 12 and Table VI.

Analysis of Loss Items: We remove each loss function and keep other settings constant in the ablation studies. Fig. 11 shows the qualitative results. 3DGS's [33] powerful rendering ability allows Gaussians to render similar images with entirely different geometry. We find that it's essential to physically constrain Gaussians' rotation and distance to achieve correct dense correspondence, especially in obstructed areas and areas with dense vertices like lips and nostrils. Although constraining the Gaussian's scale seems to have a limited impact on geometry

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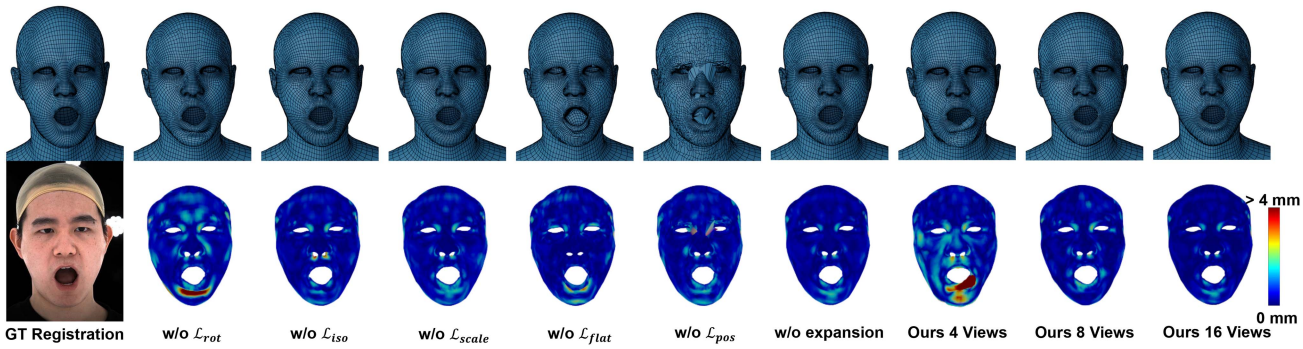


Fig. 11. Visualization of the reconstructed meshes after ablating loss items, Gaussian normal expansion operation, and view numbers. The second row displays the color-coded point-to-surface distances between the reconstructed meshes and the scans as heatmaps on the meshes' surface. **Please zoom-in for detailed observation.**

TABLE V
QUANTITATIVE COMPARISONS OF ABLATION STUDIES TOWARDS GEOMETRY RECONSTRUCTION. THE BEST AND THE SECOND-BEST RESULTS ARE MARKED WITH **BOLD** AND UNDERLINE, RESPECTIVELY. "TOPOLOGY" MEANS WHETHER IT CAN MAINTAIN THE ACCURATE TOPOLOGY.

Methods	<0.2mm(%) ↑	<2mm(%) ↑	Mean(mm) ↓	Topology
w/o $\mathcal{L}_{flat}$	21.998	**95.969**	**0.748**	✗
w/o $\mathcal{L}_{pos}$	23.192	90.678	0.852	✗
w/o $\mathcal{L}_{rot}$	17.565	87.029	1.009	✓
w/o $\mathcal{L}_{iso}$	22.510	90.420	0.904	✓
w/o $\mathcal{L}_{scale}$	22.681	91.345	0.856	✓
w/o optimization	13.188	77.008	1.502	✓
w/o blendshape	22.889	91.239	0.873	✓
w/o expansion	20.464	90.420	0.900	✓
4 views	23.022	87.440	0.979	✓
8 views	**24.727**	90.423	0.845	✓
16 views	<u>23.704</u>	<u>91.653</u>	<u>0.840</u>	✓

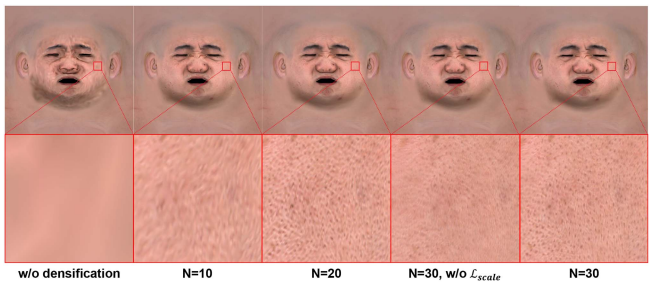


Fig. 12. Comparisons of texture quality under different settings.

TABLE VI
QUANTITATIVE COMPARISONS OF ABLATION STUDIES TOWARDS TEXTURE RECONSTRUCTION

Methods	PSNR↑	LPIPS↓	Runtime	GPU Memory
w/o dens.	26.690	0.204	19.14	6.96
N=10	28.409	0.133	20.02	7.32
N=20	28.577	0.094	23.70	8.17
N=30, w/o $\mathcal{L}_{scale}$	28.586	0.084	28.25	9.28
N=30	**28.614**	**0.083**	28.25	9.28

accuracy (an increase of around 0.016 mm), it is crucial for maintaining pore-level texture details and avoiding blurring, as shown in Fig. 12. A similar conclusion can also be drawn from the Table V that deleting scale loss and physical loss will lead to a decrease in the quantitative results. The topology prior losses $\mathcal{L}_{pos}$ and $\mathcal{L}_{flat}$ effectively maintain the stability of invisible areas, such as inner sockets, and significantly prevent exaggerated extrapolation or interpenetration. Although removing these topology preservation terms leads to the best quantitative reconstruction results, it compromises the quality of the grid and wiring, so that it cannot meet the requirements of dynamic head assets. The combination of these loss terms enables obtaining the perfectly wired and accurately reconstructed head models.

Analysis of Expression Priors: The expression prior provides relatively accurate initialization for facial expressions, especially in extreme cases. We conduct ablation studies under two settings: 1) without using blendshapes for initialization, and 2) using only blendshapes without optimization, in Table V. The first setting degenerates into vanilla Topo4D, leading to an overall decline in reconstruction quality. In the second setting, using these generalized blendshapes fails to represent personalized expressions, resulting in a significant drop in reconstruction accuracy. Only by combining blendshapes and optimization can we achieve the best reconstruction performance.

Analysis of Normal Expansion: Since we constrain the orientation and scale during initialization and optimization, Gaussians only cover a small scale in the surface normal direction. As illustrated in Fig. 11 and Table V, Gaussian normal expansion can marginally reduce the overall error but achieve more accuracy in the detailed regions where the error is less than 0.2 mm.

Analysis of Input Views: We evaluate the effect of the number of input views on mesh quality, as shown in Fig. 11 and Table V. When the views are set 4, the reconstructed mesh exhibits significant distortion. When the input views are increased to 8, it can yield competitive results with 16 views, which even achieve better results in <0.2 mm. However, the lack of some perspectives results in slightly lower global accuracy. Nevertheless, it demonstrates the applicability to capture systems with fewer cameras. The results with 16 views indicate that incorporating more views may introduce a trade-off between local and global optimization, where certain regions may fail to reach the best local alignment. However, this generally leads to superior overall reconstruction quality. In practice, the number of input views can be flexibly chosen depending on the specific requirements and capture system.

Analysis of Gaussian Density: Fig. 12 depicts the impact of different density levels on texture quality for 8K-level texture

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TABLE VII
ABLATION STUDIES OF THE BRDF CONVERSION MODULE

Methods	Albedo			Specular		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Separately	26.432	0.785	0.203	25.025	0.759	0.278
Jointly	**27.020**	**0.863**	**0.189**	**25.664**	**0.798**	**0.236**

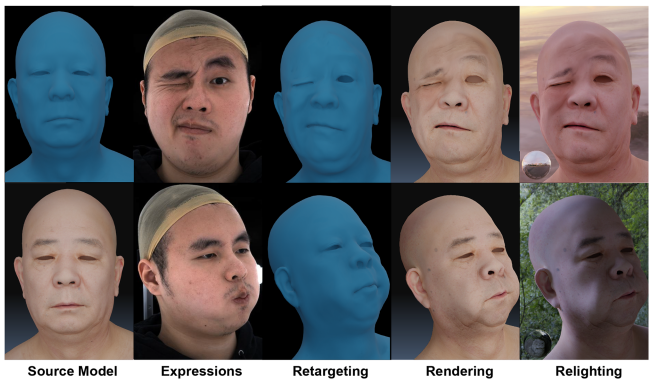


Fig. 13. We showcase retargeting results to the other face model and the relighting results.

acquisition in UV space densification. As the density decreases, the texture becomes blurred and loses detail. We also conduct quantitative experiments and efficiency comparisons across different density levels in Table VI, which are conducted on a 100-frame sequence. The quantitative results confirm our visual analysis. Although higher Gaussian density increases memory consumption and runtime, the slight overhead increase remains manageable to support practical use. Our method can adapt to arbitrary resolutions of texture, so that the Gaussian density can be tuned according to the desired texture resolution in practical use.

Analysis of BRDF Conversion Model: We compare two strategies for BRDF texture generation: predicting albedo and specular separately and jointly, on our collected individual test dataset in Table VII. In the separate setting, we train two individual networks for albedo and specular, respectively, to learn illumination removal and highlight prediction, without leveraging the interaction between the two modalities. In the joint setting, we enable the model to learn collaboratively for illumination disentanglement from albedo and specular maps. The results show that joint prediction leads to better disentanglement of albedo and specular, thereby achieving better performance.

G. Application

Driving digital characters by real performance is widely applied in industry workflows. Our method can retarget the captured expression sequence of an actor faithfully to other characters. Besides, relighting is also a widespread application in the CV and CG pipelines. Fig. 13 shows the results of retargeting extreme expressions and wrinkled texture maps learned by our methods to another facial model, as well as relighting rendering results, demonstrating that our approach integrates seamlessly into existing CG workflows, enabling direct use in production pipelines.



Fig. 14. We show some examples of failure cases.

VI. CONCLUSION

In this paper, we propose Topo4D++, an efficient framework that can extract temporally topology-consistent meshes and 8 K BRDF textures from calibrated multi-view videos. With the blendshape-based expression priors and regularizing a set of topology-aware geometrical and physical loss items, we achieve topology-preserving Gaussian optimization for faithfully capturing the subject's expressions. By densifying Gaussians in UV space, we capture realistic pore-level details at high resolution and extract high-fidelity 8 K texture maps and displacement maps. Moreover, we train a conversion model to obtain dynamic BRDF UV texture maps for physically based rendering under different light conditions. To sum up, our method provides a brand new way to reconstruct high-fidelity facial meshes and 8 K texture maps, opening up new avenues for capturing 4D digital humans in an efficient and low-cost manner.

Limitation Discussion: Topo4D++ is designed to achieve 4D facial registration without MVS reconstruction and artists' manual operation. Although our Topo4D++ is generalized and robust across different identities and different capture systems, with the ability to efficiently and automatically reconstruct 4D facial meshes with pore-level texture details, it still has some limitations. First, our method fails when heavy overlapping leads to the loss of tracking, such as sticking out the tongue as shown in Fig. 14, which is commonly solved by artists' manual operations in practical use [88]. Besides, long sequential head rotation may lead to cumulative errors, which can be addressed by applying effective tracking methods. Our approach is also constrained by relatively accurate calibration, which could be solved by large-scale pre-trained models, e.g., VGGTs [89] and colmap-free 3DGS methods [90], [91]. Finally, our method cannot directly reconstruct eyes, teeth, tongue, etc. Some methods for the inner mouth [92] and eyes [93] combined with our Topo4D++ may help resolve these challenges. We leave these issues for future work.

Ethics Discussion: In this work, all subjects have signed agreements authorizing us to use the collected data for scientific research on 4D facial reconstruction. We will make every effort to safeguard the original data from disclosure. Our method relies heavily on visual capture systems similar to the Light Stage for data collection, which can somewhat mitigate the risk of misuse. We are committed to privacy protection, preventing the misuse of 4D face reconstruction for criminal purposes.

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We have utilized the generative AI "Doubao" to help revise grammar.

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REFERENCES

[1] P. Debevec, "The light stages and their applications to photoreal digital actors," in *Proc. SIGGRAPH Asia Conf. Papers*, 2012, pp. 1–6.

[2] M. Goesele, B. Curless, and S. M. Seitz, "Multi-view stereo revisited," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2006, pp. 2402–2409.

[3] W.-C. Ma et al., "Rapid acquisition of specular and diffuse normal maps from polarized spherical gradient illumination," in *Proc. Eurographics Conf. Rendering Techn.*, 2007, pp. 183–194.

[4] P. J. Besl and N. D. McKay, "Method for registration of 3-D shapes," in *Sensor Fusion IV: Control Paradigms and Data Structures*, Bellingham, WA, USA: SPIE, 1992, pp. 586–606.

[5] Wrap4d - faceform, 2023. [Online]. Available: https://faceform.com/wrap4d/

[6] B. Gecer, S. Ploumpis, I. Kotsia, and S. Zafeiriou, "GANFIT: Generative adversarial network fitting for high fidelity 3D face reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 1155–1164.

[7] Y. Feng, H. Feng, M. J. Black, and T. Bolkart, "Learning an animatable detailed 3D face model from in-the-wild images," *ACM Trans. Graph.*, vol. 40, 2021, Art. no. 88.

[8] O. Aldrian and W. A. Smith, "Inverse rendering of faces with a 3D morphable model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 5, pp. 1080–1093, May 2013.

[9] A. Bas, W. A. Smith, T. Bolkart, and S. Wuhrer, "Fitting a 3D morphable model to edges: A comparison between hard and soft correspondences," in *Proc. Asian Conf. Comput. Vis.*, Taipei, Taiwan, 2017, pp. 377–391.

[10] S. Ploumpis et al., "Towards a complete 3D morphable model of the human head," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 11, pp. 4142–4160, Nov. 2021.

[11] T. Li, T. Bolkart, M. J. Black, H. Li, and J. Romero, "Learning a model of facial shape and expression from 4D scans," *ACM Trans. Graph.*, vol. 36, 2017, Art. no. 194.

[12] Z. Bai, Z. Cui, J. A. Rahim, X. Liu, and P. Tan, "Deep facial non-rigid multi-view stereo," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 5850–5860.

[13] L. Bao et al., "High-fidelity 3D digital human head creation from RGB-D selfies," *ACM Trans. Graph.*, vol. 41, 2021, Art. no. 3.

[14] J. Thies, M. Zollhofer, M. Stamminger, C. Theobalt, and M. Nießner, "Face2Face: Real-time face capture and reenactment of RGB videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2387–2395.

[15] E. Wood et al., "3D face reconstruction with dense landmarks," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 160–177.

[16] E. Sariyanidi, C. J. Zampella, R. T. Schultz, and B. Tunç, "Inequality-constrained 3D morphable face model fitting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 2, pp. 1305–1318, Feb. 2024.

[17] X. Zhu, C. Yu, D. Huang, Z. Lei, H. Wang, and S. Z. Li, "Beyond 3DMM: Learning to capture high-fidelity 3D face shape," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 1442–1457, Feb. 2023.

[18] A. Lattas et al., "AvatarMe: Realistically renderable 3D facial reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 757–766.

[19] A. Lattas, S. Moschoglou, S. Ploumpis, B. Gecer, A. Ghosh, and S. Zafeiriou, "AvatarMe: Facial shape and BRDF inference with photorealistic rendering-aware GANs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 9269–9284, Dec. 2022.

[20] F. Wu et al., "MVF-net: Multi-view 3D face morphable model regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 959–968.

[21] J. Guo, X. Zhu, Y. Yang, F. Yang, Z. Lei, and S. Z. Li, "Towards fast, accurate and stable 3D dense face alignment," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 152–168.

[22] S. Qian, "VHAP: Versatile head alignment with adaptive appearance priors," Sep. 2024. [Online]. Available: https://github.com/ShenhanQian/VHAP

[23] T. Li, S. Liu, T. Bolkart, J. Liu, H. Li, and Y. Zhao, "Topologically consistent multi-view face inference using volumetric sampling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 3824–3834.

[24] T. Bolkart, T. Li, and M. J. Black, "Instant multi-view head capture through learnable registration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 768–779.

[25] S. Liu, Y. Cai, H. Chen, Y. Zhou, and Y. Zhao, "Rapid face asset acquisition with recurrent feature alignment," *ACM Trans. Graph.*, vol. 41, pp. 1–17, 2022.

[26] Y. Xiao, H. Zhu, H. Yang, Z. Diao, X. Lu, and X. Cao, "Detailed facial geometry recovery from multi-view images by learning an implicit function," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 2839–2847.

[27] J. Li, D. Kang, and Z. He, "GRAPE: Generalizable and robust multi-view facial capture," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 403–418.

[28] L. Zhang et al., "Video-driven neural physically-based facial asset for production," *ACM Trans. Graph.*, vol. 41, pp. 1–16, 2022.

[29] T. Beeler et al., "High-quality passive facial performance capture using anchor frames," *ACM Trans. Graph.*, vol. 30, 2011, Art. no. 75.

[30] D. Bradley, W. Heidrich, T. Popa, and A. Sheffer, "High resolution passive facial performance capture," in *Proc. ACM SIGGRAPH*, 2010, pp. 1–10.

[31] G. Fyffe et al., "Multi-view stereo on consistent face topology," *Comput. Graph. Forum*, vol. 36, pp. 295–309, 2017.

[32] H. Zhu et al., "FaceScape: 3D facial dataset and benchmark for single-view 3D face reconstruction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14528–14545, Dec. 2023.

[33] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, 2023, Art. no. 139.

[34] J. Luiten, G. Kopanas, B. Leibe, and D. Ramanan, "Dynamic 3D Gaussians: Tracking by persistent dynamic view synthesis," 2023, *arXiv:2308.09713*.

[35] Z. Yang, X. Gao, W. Zhou, S. Jiao, Y. Zhang, and X. Jin, "Deformable 3D Gaussians for high-fidelity monocular dynamic scene reconstruction," 2023, *arXiv:2309.13101*.

[36] G. Wu et al., "4D Gaussian splatting for real-time dynamic scene rendering," 2023, *arXiv:2310.08528*.

[37] Z. Yang, H. Yang, Z. Pan, X. Zhu, and L. Zhang, "Real-time photorealistic dynamic scene representation and rendering with 4D Gaussian splatting," 2023, *arXiv:2310.10642*.

[38] Y. Liang et al., "GauFRe: Gaussian deformation fields for real-time dynamic novel view synthesis," 2023, *arXiv:2312.11458*.

[39] Agisoft metashape: Agisoft metashape, 2024. [Online]. Available: https://www.agisoft.com/

[40] C.-H. Wuu et al., "Multiface: A dataset for neural face rendering," 2022, *arXiv:2207.11243*.

[41] X. Li et al., "Topo4D: Topology-preserving Gaussian splatting for high-fidelity 4D head capture," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 128–145.

[42] V. Blanz and T. Vetter, "A morphable model for the synthesis of 3D faces," in *Proc. Annu. Conf. Comput. Graph. Interactive Techn.*, 1999, pp. 187–194.

[43] J. Booth, A. Roussos, S. Zafeiriou, A. Ponniah, and D. Dunaway, "A 3D morphable model learnt from 10,000 faces," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 5543–5552.

[44] C. Cao, Y. Weng, S. Zhou, Y. Tong, and K. Zhou, "FaceWarehouse: A 3D facial expression database for visual computing," *IEEE Trans. Vis. Comput. Graph.*, vol. 20, no. 3, pp. 413–425, Mar. 2014.

[45] P. Ji, H. Li, L. Jiang, and X. Liu, "Light-weight multi-view topology consistent facial geometry and reflectance capture," in *Proc. Comput. Graph. Int. Conf.*, 2021, pp. 139–150.

[46] J. Deng, S. Cheng, N. Xue, Y. Zhou, and S. Zafeiriou, "UV-GAN: Adversarial facial UV map completion for pose-invariant face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7093–7102.

[47] X. Ren, A. Lattas, B. Gecer, J. Deng, C. Ma, and X. Yang, "Facial geometric detail recovery via implicit representation," in *Proc. IEEE 17th Int. Conf. Autom. Face Gesture Recognit.*, 2023, pp. 1–8.

[48] R. Slossberg, I. Jubran, and R. Kimmel, "Unsupervised high-fidelity facial texture generation and reconstruction," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 212–229.

[49] L. Biwen, R. Jianqiang, F. Mengyang, C. Miaomiao, and X. Xuansong, "A hierarchical representation network for accurate and detailed face reconstruction from in-the-wild images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 394–403.

[50] A. Lattas, S. Moschoglou, S. Ploumpis, B. Gecer, J. Deng, and S. Zafeiriou, "FitMe: Deep photorealistic 3D morphable model avatars," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 8629–8640.

[51] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," *Commun. ACM*, vol. 65, pp. 99–106, 2021.

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[52] T. Müller, A. Evans, C. Schied, and A. Keller, "Instant neural graphics primitives with a multiresolution hash encoding," *ACM Trans. Graph.*, vol. 41, pp. 1–15, 2022.
[53] S. Fridovich-Keil, A. Yu, M. Tancik, Q. Chen, B. Recht, and A. Kanazawa, "Plenoxels: Radiance fields without neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5501–5510.
[54] A. Chen, Z. Xu, A. Geiger, J. Yu, and H. Su, "TensoRF: Tensorial radiance fields," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 333–350.
[55] H. Lin et al., "Efficient neural radiance fields for interactive free-viewpoint video," in *Proc. SIGGRAPH Asia Conf. Papers*, 2022, pp. 1–9.
[56] Z. Chen, T. Funkhouser, P. Hedman, and A. Tagliasacchi, "MobileNeRF: Exploiting the polygon rasterization pipeline for efficient neural field rendering on mobile architectures," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 16569–16578.
[57] S. Lombardi, T. Simon, G. Schwartz, M. Zollhoefer, Y. Sheikh, and J. Saragih, "Mixture of volumetric primitives for efficient neural rendering," *ACM Trans. Graph.*, vol. 40, pp. 1–13, 2021.
[58] P. Wang, L. Liu, Y. Liu, C. Theobalt, T. Komura, and W. Wang, "NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2021, pp. 27171–27183.
[59] L. Yariv et al., "BakedSDF: Meshing neural SDFs for real-time view synthesis," 2023, *arXiv:2302.14859*.
[60] X. Meng, W. Chen, and B. Yang, "NeAT: Learning neural implicit surfaces with arbitrary topologies from multi-view images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 248–258.
[61] J. T. Barron, B. Mildenhall, M. Tancik, P. Hedman, R. Martin-Brualla, and P. P. Srinivasan, "Mip-NeRF: A multiscale representation for anti-aliasing neural radiance fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 5855–5864.
[62] J. T. Barron, B. Mildenhall, D. Verbin, P. P. Srinivasan, and P. Hedman, "Mip-NeRF 360: Unbounded anti-aliased neural radiance fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5470–5479.
[63] Y. Jiang et al., "AligNeRF: High-fidelity neural radiance fields via alignment-aware training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 46–55.
[64] C. Gao, A. Saraf, J. Kopf, and J.-B. Huang, "Dynamic view synthesis from dynamic monocular video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 5712–5721.
[65] T. Li et al., "Neural 3D video synthesis from multi-view video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5521–5531.
[66] S. Fridovich-Keil, G. Meanti, F. R. Warburg, B. Recht, and A. Kanazawa, "K-planes: Explicit radiance fields in space, time, and appearance," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 12479–12488.
[67] Y. Yan, Z. Zhou, Z. Wang, J. Gao, and X. Yang, "DialogueNeRF: Towards realistic avatar face-to-face conversation video generation," 2022, *arXiv:2203.07931*.
[68] S. Qian, T. Kirschstein, L. Schoneveld, D. Davoli, S. Giebenhain, and M. Nießner, "GaussianAvatars: Photorealistic head avatars with rigged 3D Gaussians," 2023, *arXiv:2312.02069*.
[69] Y. Xu et al., "Gaussian head avatar: Ultra high-fidelity head avatar via dynamic Gaussians," 2023, *arXiv:2312.03029*.
[70] Y. Chen et al., "MonoGaussianAvatar: Monocular Gaussian point-based head avatar," 2023, *arXiv:2312.04558*.
[71] Z. Li, Z. Zheng, L. Wang, and Y. Liu, "Animatable Gaussians: Learning pose-dependent Gaussian maps for high-fidelity human avatar modeling," 2023, *arXiv:2311.16096*.
[72] W. Zielonka, T. Bagautdinov, S. Saito, M. Zollhöfer, J. Thies, and J. Romero, "Drivable 3D Gaussian avatars," 2023, *arXiv:2311.08581*.
[73] Z. Qian, S. Wang, M. Mihajlovic, A. Geiger, and S. Tang, "3DGS-Avatar: Animatable avatars via deformable 3D Gaussian splatting," 2023, *arXiv:2312.09228*.
[74] J. Li, Z. Kuang, Y. Zhao, M. He, K. Bladin, and H. Li, "Dynamic facial asset and rig generation from a single scan," *ACM Trans. Graph.*, vol. 39, 2020, Art. no. 215.
[75] C. Lugaresi et al., "MediaPipe: A framework for building perception pipelines," 2019, *arXiv: 1906.08172*.
[76] S. Laine, J. Hellsten, T. Karras, Y. Seol, J. Lehtinen, and T. Aila, "Modular primitives for high-performance differentiable rendering," *ACM Trans. Graph.*, vol. 39, 2020, Art. no. 194.
[77] P. Debevec, T. Hawkins, C. Tchou, H.-P. Duiker, W. Sarokin, and M. Sagar, "Acquiring the reflectance field of a human face," in *Proc. Annu. Conf. Comput. Graph. Interactive Techn.*, 2000, pp. 145–156.
[78] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 10684–10695.
[79] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2019, pp. 8026–8037.
[80] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[81] C. Loop and Z. Zhang, "Computing rectifying homographies for stereo vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 1999, pp. 125–131.
[82] Y. Lin, J. Shen, Y. Wang, and M. Pantic, "RoI tanh-polar transformer network for face parsing in the wild," *Image Vis. Comput.*, vol. 112, 2021, Art. no. 104190.
[83] C. Schuhmann et al., "LAION-5B: An open large-scale dataset for training next generation image-text models," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2022, pp. 25278–25294.
[84] B. Amberg, S. Romdhani, and T. Vetter, "Optimal step nonrigid ICP algorithms for surface registration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2007, pp. 1–8.
[85] J. L. Schonberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 4104–4113.
[86] H. Kim, M. Jang, W. Yoon, J. Lee, D. Na, and S. Woo, "SwitchLight: Co-design of physics-driven architecture and pre-training framework for human portrait relighting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 25096–25106.
[87] Z. Zeng et al., "RGB-X: Image decomposition and synthesis using material-and lighting-aware diffusion models," in *Proc. Annu. Conf. Comput. Graph. Interactive Techn.*, 2024, pp. 1–11.
[88] R. Li et al., "Learning formation of physically-based face attributes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 3410–3419.
[89] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, and D. Novotny, "VGGT: Visual geometry grounded transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2025, pp. 5294–5306.
[90] Y. Fu, S. Liu, A. Kulkarni, J. Kautz, A. A. Efros, and X. Wang, "COLMAP-free 3D Gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 20796–20805.
[91] Y. Zheng et al., "NexusGS: Sparse view synthesis with epipolar depth priors in 3D Gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2025, pp. 26800–26809.
[92] F. Rasras et al., "The lips, the teeth, the tip of the tongue: LTT tracking," in *Proc. SIGGRAPH Asia Conf. Papers*, 2024, pp. 1–11.
[93] E. Ververas, P. Gkagkos, J. Deng, M. C. Doukas, J. Guo, and S. Zafeiriou, "3DGazeNet: Generalizing 3D gaze estimation with weak-supervision from synthetic views," in *Proc. Eur. Conf. Comput. Vis.*, 2025, pp. 387–404.



Yuhao Cheng received the bachelor's degree from Shanghai Jiao Tong University, in 2019, and the master's degree from Shanghai Jiao Tong University, supervised by Prof. Peilin Liu and Prof. Fei Wen. He is currently working toward the PhD degree with the Artificial Intelligence Research Institute, Shanghai Jiao Tong University. He is supervised by Prof. Qionghai Dai and Prof. Yichao Yan. He is interested in human-centered projects. Typically, he is working on (1) 3D digital face, (2) robotic face, and (3) face perception.



Xuanchen Li received the BS degree from Jilin University, supervised by Prof. Yichao Yan, in 2024. He is currently working toward the PhD degree with the Artificial Intelligence Research Institute, Shanghai Jiao Tong University. He is interested in the reconstruction and animation of 3D digital humans.

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Xingyu Ren received the BS degree from the Harbin Institute of Technology Informatic Engineering Department, in 2016, and the PhD degree from the Artificial Intelligence Research Institute of Shanghai Jiao Tong University, in 2025. He was supervised by Prof. Chao Ma. His research interest is 3D Face Modeling and synthesis.



Jianyu Wang received the bachelor's degree from Shanghai Jiao Tong University, in 2024. He is a research assistant with the Artificial Intelligence Research Institute, Shanghai Jiao Tong University. He is supervised by Prof. Yichao Yan. He is interested in 3D digital human, optimization and reconstruction of 4D faces, as well as the reconstruction and gaze prediction of eyeballs.



Haozhe Jia received the bachelor's, master's, and PhD's degrees from Northwestern Polytechnical University, supervised by Prof. Yong Xia. He is a postdoctoral researcher with Huawei Cloud Technologies Company Ltd. His research interests include computer vision, AIGC, digital human, and deep learning.



Di Xu received the BS and PhD degrees from Nanyang Technological University, Singapore, in 2011 and 2016, respectively. He is currently a computer vision research expert in Huawei Cloud. His research interests include 3D vision, generative Al, digital human, and virtual reality.



Wenhan Zhu received the BE degree from the Huazhong University of Science and Technology, Wuhan, China, in 2015, and the PhD degree from Shanghai Jiao Tong University, Shanghai, China, in 2021. He is a post-doctoral researcher with the Artificial Intelligence Institute from 2021 to 2024. Now, he is the CEO of Xueshen AI. His current research interests are digital human applications.



Bingbing Ni received the BEng degree in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2005, and the PhD degree from the National University of Singapore, Singapore, in 2011. He is currently a professor with the Department of Electrical Engineering, Shanghai Jiao Tong University. Before that, he was a research scientist with the Advanced Digital Sciences Center, Singapore. He was with Microsoft Research Asia, Beijing, China, as a research intern, in 2009. He was also a software engineer Intern with Google Inc., Mountain View, California, in 2010. He was a recipient of the best paper award from PCM11 and the Best Student Paper Award from PREMIA08.



Yichao Yan received the BE degree from Shanghai Jiao Tong University, in 2013, the ME degree from École Centrale de Lyon in France, and the PhD degree from Shanghai Jiao Tong University, in 2019, supervised by Professor Xiaokang Yang. He is a tenure-track associate professor with Shanghai Jiao Tong University. He was a research scientist with the Inception Institute of Artificial Intelligence (IIAI) from June 2019 to April 2021. He is interested in computer vision problems. Typically, he is working on (1) 3D representation (2) AIGC (3) Embodied AI.

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